Monish H

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OBJECTIVE

B.Tech Electronics and Communication Engineering student with a diploma in Electrical and Electronics Engineering, seeking an entry-level Electrical Engineer or related technical role. Trained in electric traction power systems and transformers, with hands-on work in relay control circuits, ESP32 embedded systems, IoT sensors and circuit testing.

EDUCATION

B.Tech, Electronics and Communication Engineering 2024 - 2027

Shanmugha Arts, Science, Technology & Research Academy (SASTRA Deemed University), Thanjavur, Tamil Nadu

Diploma, Electrical and Electronics Engineering 2021 - 2024

Shanmugha Polytechnic College, SASTRA Deemed University, Thanjavur, Tamil Nadu

EXPERIENCE

Intern, Central Workshop, Golden Rock (Ponmalai), Southern Railway, Tiruchirappalli Jun 2023

- Completed industrial training on electric locomotive power systems, covering how traction current is received, transformers, onboard batteries and current graphs.
- Observed electrical systems and processes in a working railway maintenance workshop.

PROJECTS

IoT Smart Home Automation Diploma Project, 2023 - 2024

- Built and tested an ESP32 home automation circuit with an 8-channel relay module, switching 4 lamps and 4 plug points.
- Gave each load four control options: manual switch, IR remote (TSOP1838 receiver), Google Assistant voice commands and the ESP RainMaker and Google Home apps.
- Sent DHT11 temperature, humidity and LDR light readings to the mobile app every 2 seconds, with live device status.
- Kept manual switch and IR control working during Wi-Fi outages. Firmware written in Arduino C++.

AI-Based Helmet Detection System Using YOLOv8 B.Tech Mini Project

- Trained a YOLOv8 model on three annotated Roboflow datasets to classify riders as Helmet or No Helmet (50 epochs, image size 640, batch size 16).
- Ran real-time detection on images and webcam video using Python and OpenCV, with bounding boxes, class labels and confidence scores on screen.
- Runs fully offline on a local machine and can be integrated with CCTV systems for traffic monitoring.

TECHNICAL SKILLS

Electrical: Transformers, relays, circuit wiring and testing, electric traction basics
Tools: AutoCAD, MATLAB, Arduino IDE
Programming: C, C++, Python
Embedded and IoT: ESP32, sensors (DHT11, LDR), IR receivers, Internet of Things (IoT)
Computer Vision: YOLOv8, OpenCV, Roboflow
Professional: Team leadership, coordination, data management, logistics

CERTIFICATIONS

Signal Processing Onramp, MathWorks (MATLAB) Jul 2025

ACHIEVEMENTS

Certificate of Merit, ranked 2nd in the department in the 3rd and 5th semesters of the diploma.

LEADERSHIP & ACTIVITIES

Head, KS Merchandise, Kuruksastra 2025 - Present

- Joined the club in 2025 and became Head for the 2026-27 academic year.
- Lead a club of 70+ members that pitches and sells merchandise such as tees and glow bands to students for Kuruksastra, SASTRA's cultural fest.
- Oversee members' work across data collection, data management and data logistics.

Marketing Cluster Member, Colosseum Jun 2025 - Feb 2026

- Promoted Colosseum, SASTRA's sports fest.

Member, Content Creation, KS Public Relations 2025 - 2026

- Helped create content for the club's social media pages.